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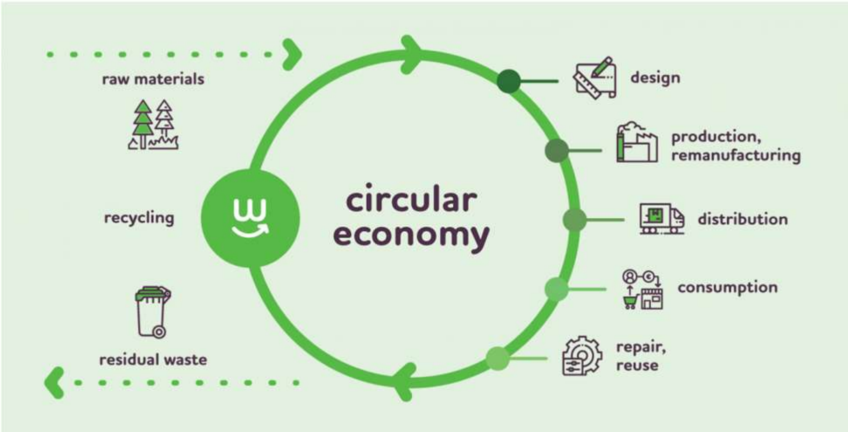
Digital
Product
Passport

Introduction

Digital product passports (DPP) aim to gather data on a product and its supply chain and share it across entire value chains so all actors, including consumers, have a better understanding of the materials and products they use and their embodied environmental impact.

The digital product passports initiative is part of the proposed Ecodesign for Sustainable Products Regulation and one of the key actions under the Circular Economy Action Plan (CEAP). The goal of this initiative is to lay the groundwork for a gradual introduction of a digital product passport in at least three key markets by 2024. These include textiles, construction, industrial and electric vehicle batteries, and at least one other of the key value chains identified in the Circular Economy Action Plan such as consumer electronics, packaging, and food



DPPs support the principles of the circular economy by promoting reuse, recycling, and responsible disposal of products. The digital passport can contain information about the recyclability of materials, facilitating better resource management.

DPPs aim to provide consumers, businesses, and regulators with transparent and easily accessible information about a product's origin, materials, manufacturing processes, and environmental impact. This transparency is crucial for informed decision-making and fostering trust.

Key dates for fashion

- ✓ **2026/2027:** final EU DPP regulations confirmed
- ✓ **2030:** all items in the supply chain and stores should have a digital product passpost.

What is a Digital Product Passport

A Digital Product Passport (DPP) is a digital representation that contains comprehensive information about a specific product. While the specific details may vary based on industry and product type, here are some common types of information that can be included in a Digital Product Passport, especially in the context of consumer goods and apparel:

1.Product Identification:
- ✓ Unique product identifier or code.
- ✓ Product name and description.

2.Design and Specifications:
- ✓ 3D models or images of the product.
- ✓ Design specifications and technical details.
- ✓ Information about components and materials used in the product.

3.Materials Information:
- ✓ Details about the origin and composition of materials.
- ✓ Certifications and standards associated with materials (e.g., organic, recycled, fair trade).

4.Supply Chain Information:
- ✓ Information about suppliers and manufacturers involved in the production process.
- ✓ Geographic locations of suppliers.

5.Manufacturing Processes:
- ✓ Details about the methods and processes used in manufacturing.
- ✓ Energy consumption and emissions during production.

6. **Environmental Impact:**
   - ✓ Carbon footprint of the product.
   - ✓ Water usage and other environmental metrics.
   - ✓ Life cycle assessment data.

7. **Sustainability Certifications:**
   - ✓ Any certifications or labels indicating the product's sustainability or eco-friendliness.

8. **End-of-Life Information:**
   - ✓ Guidance on product disposal and recycling.
   - ✓ Information about recyclability and biodegradability.

9. **User Instructions:**
   - ✓ Care instructions for the product.
   - ✓ Guidance on maintenance and repair.

10. **Digital Connectivity:**
   - ✓ Integration with digital platforms for updates and information retrieval.
   - ✓ QR codes or NFC tags for quick access to the digital passport.

11. **Blockchain or Distributed Ledger Information:**
   - ✓ If applicable, information related to the use of blockchain for securing and validating data.

12. **Consumer Engagement:**
   - ✓ Features that allow consumers to access the digital passport for transparency and information.
   - ✓ Educational content about the product's sustainability and ethical considerations.

Example Digital Product Passport

Allshoes safety footwear

Environmental impact: Redbrick Pearl

Product(ion)

Materials

	gr	% recycled
Steel	18	
Glue solvent-based	25	
Plastisol	2	
Polyester	274	48% (avg)
PET	14	100%
R-PES (35%) 100% recycled + PU + Acrylic Resin	138	36%
PU (49%) 100% recycled, Polyester (10%) 100% recycled, PU (20%), castor oil (1%), binder (15%), carbon (5%)	62	59%
Polyester / Polypropylene / Resin	6	
TPU	387	84% (avg)
PU	248	35%
Polyamide	124	47%
Polyester (50%) 100% recycled / Polycaprolactone (50%)	11	50%

Recycled materials (weight-based) 53%

Genoemde gerecyclede percentages zijn een momentopname. We streven ernaar om dit te blijven verhogen.

Number of different materials used 12

Production location 1st Tier* supplier(s) Portugal

Production location 2nd Tier supplier(s)

Italy, Spain, Portugal, Germany, Spain, Netherlands

*Tier 1 suppliers are the direct suppliers of Allshoes Safety Footwear. Tier 2 suppliers are our suppliers' suppliers.

Use

User instructions

For longevity of the product, let product breathe after wearing and use recommended care products.

Repair instructions

Repairs might affect certification on safety standards: ISO20345:2022

Spare part/ accessoires availability

Use recommended accessoires, available on www.redbrick.eu.

End-of-life

End-of-life instructions

At the end-of-life, do not discard product as waste. Hand in the product at a collection point of the Circular Footwear Alliance so it can be recycled. See here for more information: https://www.cfalliance.eu/en/.

Recyclability

Recycling of materials possible by mechanical shredding.

End-of-life packacing

Shoebox is made from 100% recycled FSC cardboard, with waterbased ink. Please discard with paper waste.

June 2023 Version 1.6

Allshoes safety footwear

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Environmental impact: Redbrick Pearl

Environmental impact (LCA)

Scope

The production of 1 pair of the Redbrick Pearl in size 42, excluding the shoebox, including the transport from the factory to the Allshoes Warehouse.

Stages

A1 (Materials); A2 (Transport); A3 (Production); A4 (Transport gate to site)

CO2-equivalent 10.12 kg CO2-eq

LCA Consultant Ecochain

LCA verified by third party

LCA report validated by Ecoreview.

Full LCA report *Available on request*

LCA validation document of proof

EcoReview

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Ruben van Gaalen
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Date, 19-5-2023

Subject: Verification LCA for AMF and Grisport safety footwear by Allshoes

Dear Anne Jacobs,

I hereby confirm that, following detailed examinaion as independent 3rd party verifier, I have not been able to trace any relevant deviations in the LCA and by its project report from the requirements outlined in the corresponding product category regulations based on ISO 14040 and ISO 14044. The company-specific data have been examined as regards plausibility and consistency; the declaration owner is responsible for its factual integrity.

The verification has been performed for the following reports:

- LCA background report - Allshoes Safety Footwear - Supplier: AMF - Safety shoes: Onyx, Topaz, Pearl, Tourmaline & Opal
- LCA background report - Allshoes Safety Footwear - Supplier: Grisport - Safety shoes: Grisport 803 & Grisport 103 *(50% recycled content, weight-based)

Authored by Anne Jacobs from Allshoes and Aniek Baltussen from Ecochain.

The results (kg CO2 eq) of the LCAs performed are presented in the table below. For module A4 only transport is calculated from suppliers to the warehouse of Allshoes in Alkmaar. Transport to final customer still needs to be calculated.

Results tabel A1-A3 + A4

	Climate Change Kg CO2 eq				
Product	*A1*	*A2*	*A3*	*A4*	*A1-A4*
AMF – Onyx					
AMF – Topaz					
AMF – Pearl	0,30	9,39	0,08	0,35	10,12
AMF – Tourmaline	0,27	8,36	0,08	0,32	9,02
AMF – Opal	0,27	8,81	0,08	0,33	9,49
Grisport – 803	[redacted]				
Grisport – 103	[redacted]				

Ruben van Gaalen is registered on behalf of EcoReview at NMD (Foundation for the Dutch National Environmental Database) as an Accredited LCA expert. He is hereby authorized by NMD to perform this verification. Results of the LCA are a self-declaration by the producer and therefore remain under the responsbility of the producer.

Sincerely,
Ruben van Gaalen

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Allshoes calculated the **environmental impact** of the production of 1 pair of the Redbrick Pearl in size 42, excluding the shoebox, including the transport from the factory to Allshoes Warehouse.

10.12kg CO2-eq

93% Materials

3% Energy

4% Transport

June 2023 Version 1.6

DPP benefits the entire fashion value chain

Raw material supplier

- Drives trust in sourcing methods via transparency
- Facilitates raw material recovery for remanufacturing
- Substantiates recycled material composition in raw materials

Manufacturer

- Delivers post-sale data and insights from customers
- Facilitates warranty claims and recalls
- Thwarts product piracy and counterfeiting
- Substantiates sustainability claims

Retailer

- Provides product identification
- Enables access to essential product information
- Ensures authenticity of product
- Supplies customers with trusted information

Consumer

- Allows product comparison by sustainability attributes
- Delivers access to care, maintenance, and usage instructions
- Provides services related to garment, like maintenance, repair
- Locates recycling services

Repairer/ Recycler

- Provides trusted information on repair
- Identifies preventative repair opportunities
- Enables automated sorting of materials
- Delivers access to info for maintenance, repair or upgrade

Why Digital Product Passport?

1.Transparency: DPPs aim to provide consumers, businesses, and regulators with transparent and easily accessible information about a product's origin, materials, manufacturing processes, and environmental impact. This transparency is crucial for informed decision-making and fostering trust.

2.Circular Economy: DPPs support the principles of the circular economy by promoting reuse, recycling, and responsible disposal of products. The digital passport can contain information about the recyclability of materials, facilitating better resource management.

3.Supply Chain Visibility: Digital Product Passports contribute to better visibility and traceability in supply chains. This is especially important for industries where understanding the origin and lifecycle of products is critical, such as in the food and electronics sectors. In combination with RFID

4.Market Differentiation: Companies that adopt DPPs may use them as a tool for market differentiation. By showcasing the sustainability and transparency of their products, businesses can appeal to consumers who prioritize ethical and environmentally friendly choices.

5.Data-driven Decision Making: DPPs provide a structured and standardized way of collecting and sharing data about products. This data can be analyzed to identify opportunities for improvement in design, manufacturing processes, and environmental impact, leading to more informed decision-making.

Who should access DPP related info?

Access to Digital Product Passport (DPP) information is a critical consideration to balance transparency, accountability, and privacy concerns. The stakeholders who should be able to access DPP-related information may include:

1. **Consumers:** Consumers are primary stakeholders who may want access to DPP information to make informed purchasing decisions. This includes details about a product's origin, materials used, environmental impact, and other relevant information.

2. **Businesses and Supply Chain Partners:** Manufacturers, suppliers, and other entities in the supply chain need access to DPP information to ensure compliance with regulations, manage the supply chain effectively, and make informed decisions about sourcing materials.

3. **Regulators and Authorities:** Government agencies and regulatory bodies may require access to DPP information to enforce compliance with standards, regulations, and laws related to product safety, environmental impact, and other relevant criteria.

4. **Recyclers and Waste Management Authorities:** Entities involved in recycling and waste management need access to DPP information to understand the composition of products, facilitating proper recycling processes and waste disposal.

5. **Product Designers and Engineers:** Those involved in product design and engineering may require access to DPP information to improve product design, identify opportunities for innovation, and enhance the sustainability of products.

6. **Auditors and Certification Bodies:**
   Independent auditors and certification bodies may need access to DPP information to verify compliance with standards and certifications related to product quality, safety, and environmental impact.
7. **Internal Stakeholders:**
   Within a company, various departments and personnel may need access to DPP information based on their roles. This could include marketing, legal, and research and development teams.

Steps to implement DPP

Setting up a Digital Product Passport (DPP) in the apparel industry involves several specific steps tailored to the characteristics of clothing and the fashion supply chain. Here are steps to guide you in setting up a DPP in the apparel sector:

1. **Define Objectives and Scope:** Clearly define the goals of implementing a DPP in the apparel industry. Consider objectives such as enhancing transparency, improving sustainability, and providing consumers with detailed product information.

2. **Identify Key Information for Apparel:** Determine the specific information relevant to apparel products, such as material details (fabric type, origin), design specifications, manufacturing processes, ethical and sustainability aspects, and care instructions.

3. **Select DPP Technology:** Choose the technology and platform for creating and managing the DPP. Consider solutions that support digital representation, data storage, and interoperability within the apparel supply chain.
4. **Collaborate with Suppliers and Manufacturers:** Engage with suppliers and manufacturers to gather information about the materials used, production processes, and other relevant details. Collaboration is essential for obtaining accurate and up-to-date information.

5. **Standardize Size and Fit Information:** If applicable, standardize size and fit information to ensure consistency across different apparel products. This can enhance the consumer experience and reduce returns due to sizing issues.

6. **Integrate with Design Tools and PLM Systems:** Integrate the DPP system with design tools and Product Lifecycle Management (PLM) systems used in the apparel industry. This ensures seamless data transfer between the design phase and the digital representation.

7. **Include Sustainability Metrics:** Incorporate sustainability metrics, such as carbon footprint, water usage, and adherence to ethical labor practices. Provide information about certifications related to sustainable and eco-friendly practices.

8. **Address Data Security:** Implement robust data security measures to protect sensitive information in the DPP. Consider encryption, access controls, and compliance with data privacy regulations.

9. **Create User-Friendly Interface:** Design a user-friendly interface for stakeholders to access and update the DPP. Consider incorporating visuals, QR codes, or other features to enhance accessibility.

10. **Pilot Program and Test:** Conduct a pilot program with a subset of apparel products to test the functionality and gather feedback. Use this feedback to make necessary improvements before a full-scale implementation.

11. **Train Stakeholders:** Provide training to internal teams, suppliers, and other stakeholders on how to use and contribute to the DPP. Clear communication is crucial for successful implementation.

12. **Launch and Communicate:** Launch the DPP and communicate its presence to relevant stakeholders. Consider how consumers will access information—this could involve QR codes on labels, online platforms, or other methods.

13. **Continuous Monitoring and Improvement:** Continuously monitor the performance of the DPP. Gather feedback from users and stakeholders and make improvements based on evolving product information or changes in the industry.

14. **Comply with Industry Regulations:** Ensure that the DPP complies with industry regulations, standards, and any emerging guidelines related to transparency and sustainability in the apparel sector.

Setting up a DPP in the apparel industry requires collaboration among various stakeholders, including designers, manufacturers, suppliers, and retailers. It aims to provide consumers with detailed insights into the products they purchase, fostering transparency and sustainability in the fashion supply chain.

How to store DPP related data?

Storing Digital Product Passport (DPP) related information involves creating a centralized repository where comprehensive data about each product can be securely stored, managed, and accessed. Here are steps to effectively store DPP related information:

1. **Choose a Data Storage Solution:** Select a suitable data storage solution based on your organization's needs, scalability, and security requirements. Options include databases, cloud storage services, and specialized Product Lifecycle Management (PLM) systems.

2. **Design a Data Schema:** Define a structured data schema that outlines the attributes and fields to be captured for each product in the DPP. This schema should accommodate various types of information, such as product specifications, materials data, sustainability metrics, and supply chain details.

3. **Capture and Input Data:** Collect and input relevant data into the storage system. This may involve collaborating with suppliers, manufacturers, and other stakeholders to gather accurate and up-to-date information about each product.

4. **Standardize Data Formats:** Standardize data formats and terminology to ensure consistency across different products and suppliers. This facilitates easier data exchange, analysis, and reporting.

5. **Ensure Data Security:** Implement robust data security measures to protect the integrity and confidentiality of the stored information. This may include encryption, access controls, and regular security audits.

7. **Integrate with Existing Systems:** Integrate the DPP storage system with existing enterprise systems, such as PLM, Enterprise Resource Planning (ERP), and Customer Relationship Management (CRM) systems. This ensures seamless data transfer and interoperability across different departments.

8. **Facilitate Accessibility:** Design a user-friendly interface that allows authorized stakeholders to easily access and retrieve DPP related information. Consider implementing search functionality, filters, and customizable views to enhance usability.

9. **Provide Version Control:** Implement version control mechanisms to track changes and revisions to DPP information over time. This ensures that stakeholders have access to the most up-to-date and accurate data.

10. **Backup and Disaster Recovery:** Establish backup and disaster recovery procedures to prevent data loss and ensure business continuity. Regularly backup DPP data and store copies in secure locations.

11. **Comply with Regulations:** Ensure compliance with relevant regulations and data privacy laws when storing DPP related information. This may include GDPR in Europe or other regional data protection regulations.

12. **Monitor and Audit Data:** Regularly monitor and audit the stored data to ensure accuracy, completeness, and compliance with internal policies and external regulations.

13. **Continuous Improvement:** Continuously evaluate and optimize the storage system based on feedback from users and stakeholders. Identify areas for improvement and implement enhancements to enhance the effectiveness and efficiency of storing DPP related information.

Added Value Digital Product Passport

Apart from DPP EU regulations - a digital product passport can offer several added values:

1. **Traceability:** It provides a transparent record of the product's lifecycle, including its manufacturing process, materials used, and any associated data like environmental impact or certifications. This traceability can enhance trust between consumers and producers.
2. **Product Authentication**: It allows consumers to verify the authenticity of the product, ensuring that they are purchasing genuine goods and not counterfeit items.
3. **Supply Chain Management**: By digitizing product information, it becomes easier to manage the supply chain, track inventory, and optimize production processes. This can lead to cost savings and operational efficiencies – in combination with RFID.
4. **Consumer Empowerment**: Consumers can make more informed purchasing decisions by accessing detailed information about the product's origin, composition, and sustainability credentials. This aligns with growing consumer demand for transparency and ethical consumption.
5. **Sustainability and Environmental Impact**: Digital product passports can include data on a product's environmental footprint, such as its carbon footprint, energy usage, and recycling instructions. This information enables consumers to choose products with lower environmental impact and encourages companies to improve their sustainability practices.
6. **Warranty and Support**: It can serve as a digital record for warranties, user manuals, and support information, making it easier for consumers to access assistance when needed.

Overall, a digital product passport adds value by enhancing transparency, trust, and sustainability throughout the product lifecycle, benefiting both consumers and producers.

From Packaging Supplier Perspective

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1. **Packaging and Labels:** Packaging and labels plays a crucial role as the carrier of this information and a key part of how the consumer can access this information pre and post sale.

2. **Unique Code Carrier:** Both NFC and QR Codes can carry a unique ID which refers to DPP related information for particular product.

3. **QR Codes Technology:**
   - ✓ QR Codes might be preferred since the majority of end consumers are already used to QR Codes to access additional product information.
   - ✓ QR codes must be embedded in eg a printed fabric label or a woven label so that it is permanently attached to the product. The product must be traceable throughout its existence.

4. **NFC Technology:** NFC technology is also a means by which the customer can request relevant product information such as DPP-related information.

5. **DPP related packaging:** select or develop the right product that is attached to the item and lasts the entire life cycle of the item.

6. **Unique Code:** DPP requires a unique code to be encoded in the QR Code/NFC inlay. Unique code system integration with brand is required

DPP QR Labels roadmap

Applying Digital Product Passport (DPP) QR labels to apparel items involves integrating digital tracking and information systems with physical product labeling. Here's how you can do it:

1. **Digital Product Passport Integration**: Set up a digital platform or database where you can store comprehensive information about each apparel item, including details about materials, manufacturing processes, supply chain transparency, and sustainability credentials. Ensure that this system is capable of generating unique QR codes for each item.
2. **Generate DPP QR Codes**: Generate unique QR codes for each apparel item in your digital product passport system. These QR codes should link directly to the corresponding product information stored in your database.
3. **Design and Print Labels**: Design labels that include the DPP QR code along with any relevant branding or product information. Consider using durable materials and printing techniques that can withstand washing and wear.
4. **Label Placement**: Decide where to place the DPP QR code labels on the apparel/footwear items. Common locations include hangtags, care labels, or packaging. Ensure that the labels are easily visible and accessible to consumers. "How to apply" is subject to item life cycle regulation.
5. **Label Attachment**: Attach the DPP QR code labels securely to the apparel/footwear items using appropriate methods.
6. **Testing and Quality Control**: Test the scannability of the DPP QR codes using various smartphones and scanning apps to ensure that they link to the correct product information. Conduct quality control checks to verify the accuracy of the information encoded in the QR codes.

7. **User Instructions**: Provide clear instructions to consumers on how to scan the DPP QR codes using their smartphones. Include information on compatible scanning apps and any special considerations for scanning, such as lighting conditions or camera settings.
8. **Promotion and Education**: Promote the use of DPP QR codes as a means for consumers to access detailed information about the apparel items, including their sustainability and ethical credentials. Educate consumers about the benefits of scanning the codes and making informed purchasing decisions.

QR Code vs NFC

The choice between using QR codes or NFC for unique code carriers depends on several factors, including the specific requirements of your application, user experience considerations, and technical capabilities.

Here's a comparison of when each technology might be more suitable.

Use QR Codes When:

1. **Cost-Effectiveness:** QR codes are generally more cost-effective to produce and implement compared to NFC tags. They can be printed on various surfaces, such as packaging or labels, without requiring specialized materials.
2. **Wide Compatibility:** QR codes can be scanned by most smartphones with a built-in camera and do not require additional hardware. This broad compatibility makes QR codes accessible to a wide range of users without the need for specialized devices.
3. **Information Display:** QR codes can encode various types of information, such as URLs, text, or contact information. They are suitable for linking users to websites, digital content, or providing detailed product information.
4. **Marketing and Engagement:** QR codes can be used for marketing and engagement purposes, such as providing access to promotional offers, loyalty programs, or interactive experiences.

Use NFC When:

1. **Contactless Interaction:** NFC enables contactless interaction between devices at close range (typically a few centimeters). This makes it suitable for applications where users need to quickly and securely access information or perform transactions.
2. **Embedded Applications:** NFC tags can be embedded within products or integrated into packaging, enabling seamless and discreet interaction without the need for visible labels or markers.
3. **Data Storage and Updates:** NFC tags have the capacity to store more data compared to QR codes and can be updated dynamically. This makes them suitable for applications requiring frequent updates or storing large amounts of information, such as product authentication or inventory tracking.
4. **Mobile Payments and Transactions:** NFC technology is commonly used for mobile payments and contactless transactions, enabling users to make secure payments using their smartphones or other NFC-enabled devices.

Ultimately, the choice between QR codes and NFC depends on the specific use case, considering factors such as cost, compatibility, security, and user experience requirements. In some cases, a combination of both technologies may be used to provide users with flexibility and convenience in accessing information or interacting with products.